WORKED SOLUTION

STEP II 1997, Question 3

Pure • Partial Fractions Integration

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s2-q3

The key idea

The two quadratics in the denominators multiply to give $x^4 + 4$. That's the classic "difference of two squares in disguise": $x^4 + 4 = (x^2 + 2)^2 - (2x)^2$. Once we have the partial fractions, each piece is a standard integral. We split each numerator into a multiple of the derivative of its denominator, which integrates to a logarithm, plus a constant, which gives an arctangent after completing the square.

Solution

The partial fractions

First note that

$$(x^2 + 2x + 2)(x^2 - 2x + 2) = (x^2 + 2)^2 - (2x)^2 = x^4 + 4x^2 + 4 - 4x^2 = x^4 + 4.$$

So, multiplying the required identity through by $x^4 + 4$, we need

$$(ax + b)(x^2 - 2x + 2) + (cx + d)(x^2 + 2x + 2) = 1 \quad \text{for all } x.$$

Expanding the left-hand side gives

$$(a + c)x^3 + (b - 2a + d + 2c)x^2 + (2a - 2b + 2c + 2d)x + (2b + 2d).$$

Comparing coefficients with the constant polynomial 1:

$$\begin{aligned} x^3: &\quad a + c = 0, \\ x^2: &\quad b + d - 2a + 2c = 0, \\ x^1: &\quad 2(a + c) + 2(d - b) = 0, \\ x^0: &\quad 2(b + d) = 1. \end{aligned}$$

The first and third equations give $c = -a$ and $d = b$. The fourth then gives $b = d = \frac{1}{4}$. Finally, the second becomes $\frac{1}{2} - 4a = 0$, so $a = \frac{1}{8}$. Therefore

$$a = \tfrac{1}{8}, \qquad b = \tfrac{1}{4}, \qquad c = -\tfrac{1}{8}, \qquad d = \tfrac{1}{4},$$

that is,

$$\frac{1}{x^4+4} = \frac{1}{8}\left(\frac{x+2}{x^2+2x+2} + \frac{2-x}{x^2-2x+2}\right).$$

(We could have predicted $c = -a$ and $d = b$ without any algebra. Replacing x by $-x$ leaves $\frac{1}{x^4+4}$ unchanged and swaps the two denominators.)

The integral

Ask yourself How do we integrate $\dfrac{x+2}{x^2+2x+2}$?

The derivative of the denominator is $2x+2$. So we write the numerator as a multiple of $2x+2$ plus whatever is left over:

$$x+2 = \tfrac{1}{2}(2x+2)+1, \qquad 2-x = -\tfrac{1}{2}(2x-2)+1.$$

Completing the square, $x^2 \pm 2x + 2 = (x \pm 1)^2 + 1$. Hence

$$\frac{1}{x^4+4} = \frac{1}{16}\cdot\frac{2x+2}{x^2+2x+2} - \frac{1}{16}\cdot\frac{2x-2}{x^2-2x+2} + \frac{1}{8}\cdot\frac{1}{(x+1)^2+1} + \frac{1}{8}\cdot\frac{1}{(x-1)^2+1}.$$

Both quadratics are positive for all real x, so every term is continuous on $[0,1]$. The first two terms have the form f'/f with $f > 0$ and integrate to $\ln f$. The last two integrate to arctangents. So

$$\begin{aligned}
\int_0^1 \frac{\mathrm{d}x}{x^4+4} &= \frac{1}{16}\Big[\ln(x^2+2x+2) - \ln(x^2-2x+2)\Big]_0^1 + \frac{1}{8}\Big[\tan^{-1}(x+1) + \tan^{-1}(x-1)\Big]_0^1 \\
&= \frac{1}{16}[(\ln 5 - \ln 1) - (\ln 2 - \ln 2)] \\
&\quad + \frac{1}{8}[(\tan^{-1} 2 + \tan^{-1} 0) - (\tan^{-1} 1 + \tan^{-1}(-1))] \\
&= \frac{1}{16}\ln 5 + \frac{1}{8}[\tan^{-1} 2 - (\tfrac{\pi}{4} - \tfrac{\pi}{4})] \\
&= \frac{1}{16}\ln 5 + \frac{1}{8}\tan^{-1} 2,
\end{aligned}$$

as required. The $\ln 2$ terms and the $\frac{\pi}{4}$ terms cancel in pairs, which is a pleasing consequence of the symmetry we noticed above.

Checks

- At $x = 0$ the partial fractions give $\frac{1}{8}(\frac{2}{2} + \frac{2}{2}) = \frac{1}{4} = \frac{1}{0+4}$. At $x = 1$ they give $\frac{1}{8}(\frac{3}{5} + \frac{1}{1}) = \frac{1}{5}$, which is also correct.
- The integrand decreases from $\frac{1}{4}$ to $\frac{1}{5}$ on $[0,1]$, so the integral must lie between 0.2 and 0.25. Numerically, $\frac{1}{16}\ln 5 + \frac{1}{8}\tan^{-1} 2 \approx 0.100589 + 0.138394 = 0.238983$. Simpson's rule on the integral gives 0.2389834593, in agreement to about 16 significant figures.

Common slips

- Trying to factorise $x^4 + 4$ over the reals by looking for real roots. It has none, so the factors must be quadratics. Adding and subtracting $4x^2$ to complete the square is the trick.
- Forgetting the lower limit of the arctangent terms. $\tan^{-1}(0+1) = \frac{\pi}{4}$ and $\tan^{-1}(0-1) = -\frac{\pi}{4}$ only cancel when both are included.

Beyond STEP

The factorisation $x^4 + 4 = (x^2 + 2x + 2)(x^2 - 2x + 2)$ is Sophie Germain's identity, $a^4 + 4b^4 = (a^2 + 2ab + 2b^2)(a^2 - 2ab + 2b^2)$. It shows, for example, that $n^4 + 4$ is prime only when $n = 1$. More generally, every real polynomial factorises into real linear and quadratic factors, because complex roots come in conjugate pairs. So partial fractions, and hence logs and arctangents, are enough to integrate any rational function. With contour integration you can evaluate $\int_{-\infty}^{\infty} \frac{\mathrm{d}x}{x^4+4} = \frac{\pi}{4}$ almost without computation; Needham's *Visual Complex Analysis* explains why.

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